# ECON 381: Practice Midterm 2

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Name: _____

## Instructions

1. Write your name on this exam, your note sheet, and every answer sheet.
2. **Do not open this exam until instructed to do so.**
3. Silence your cell phone and place it in your bag, not in your pocket.
4. This exam is worth 100 points, with 5 additional bonus points available. The maximum recorded score is 100.
5. Check that you have all 7 pages. You have 75 minutes to complete the exam.
6. You may use one $8.5 \times 11$ sheet of handwritten notes, on both sides, prepared before the exam. Turn in your note sheet with your exam and answer sheets.
7. Write all answers and supporting work on separate blank sheets of paper. Do not write answers on the exam pages. Clearly number each answer page, label each answer with its section, question number, and part. For each multiple-choice question, write the question number and the corresponding one answer letter.
8. Use other blank sheets for scratch work. Clearly cross out any work you do not want graded in your answer sheets.
9. One standalone, non-programmable calculator is allowed. All other electronic devices are prohibited. For confidence intervals, use the critical values provided in the questions.
10. Sign the honor pledge below. At the end of the exam, turn in this exam, your answer sheets, and your note sheet.
11. **Good luck!**

**I pledge on my honor as a University of Tennessee student that I will neither give nor receive aid on this examination.**

Signature: _____

## 1. Multiple Choice Questions (30 points total; 10 questions; 3 points each)

**Instructions:** On your separate answer sheets, write each question number and **one** answer letter (a, b, c, or d). No explanation is required. Selecting more than one answer earns no credit for that question.

1. In a regression of hourly wage on years of education, which statement is least-squares assumption A1?

   (a) The sample residuals are all positive.
   (b) $E[u_i \mid \text{Education}_i = x] = 0$ at every education level $x$.
   (c) Every worker has $u_i = 0$.
   (d) $\text{Var}(u_i \mid \text{Education}_i)$ is constant.

2. Suppose ability raises wages and is systematically higher among workers with more education. Ability is part of $u_i$ in a proposed causal wage–education model, so A1 fails. Drawing a much larger random sample

   (a) eliminates ability from the regression error.
   (b) guarantees that the OLS slope equals the causal effect.
   (c) reduces sampling uncertainty but does not repair this failure of A1.
   (d) guarantees that A1 becomes true.

3. Several sampled districts share county policies and local economic shocks, making their observations dependent. Which least-squares assumption is directly threatened?

   (a) The observations are independent and identically distributed.
   (b) The conditional mean of the regression error is zero.
   (c) Unusually large outliers are unlikely.
   (d) Homoskedasticity: the conditional error variance is constant.

4. Assume A1–A3 and homoskedasticity. If sample size increases from $n$ to $4n$, holding the population error variance and regressor variance fixed, the large-sample variance and standard deviation of the OLS slope become, respectively,

   (a) unchanged and one-half of their original values.
   (b) four times and twice their original values.
   (c) one-half and one-quarter of their original values.
   (d) one-quarter and one-half of their original values.

5. Under A1–A3 and homoskedasticity, compare two wage–education studies with the same $n$. In the second population, $\text{Var}(u)$ is four times as large and Var(Education) is twice as large. The large-sample slope variance is

   (a) twice as large, and its standard deviation is $\sqrt{2}$ times as large.
   (b) one-half as large, and its standard deviation is one-half as large.
   (c) four times as large, and its standard deviation is twice as large.
   (d) unchanged, and its standard deviation is unchanged.

6. A researcher tests $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ in a test-score–income regression. A Type-I error occurs when the researcher

   (a) reports a positive slope estimate.
   (b) rejects $H_0$ even though the population slope is zero.
   (c) does not reject $H_0$ even though the population slope is zero.
   (d) rejects $H_0$ when the population slope is nonzero.

7. For a two-sided 5% test of $H_0 : \beta_1 = 0$, which set contains the three equivalent rejection rules? Use the same standard error and large-sample normal approximation throughout.

   (a) $t > 1.96$; $p < 0.05$; the 90% CI excludes zero.
   (b) $|t| < 1.96$; $p > 0.05$; the 95% CI includes zero.
   (c) $|t| > 1.96$; $p < 0.05$; the 95% CI excludes zero.
   (d) $|t| > 1.65$; $p < 0.05$; the 99% CI excludes zero.

8. A large-sample two-sided test of a zero wage–education slope gives $t = -1.80$ and $p = 0.072$. Which decision is correct?

   (a) Reject at 10%, but do not reject at 5%.
   (b) Reject at both the 10% and 5% significance levels.
   (c) Do not reject at either level because $t$ is negative.
   (d) Do not reject at 10%, but reject at 5%.

9. Let $D_i = 1$ for districts with $STR_i \geq 20$ and $D_i = 0$ otherwise. Under A1, suppose $E[\text{TestScore}_i \mid D_i = 0] = 660$ and $E[\text{TestScore}_i \mid D_i = 1] = 652$. In the regression $\text{TestScore}_i = \beta_0 + \beta_1 D_i + u_i$,

   (a) $\beta_0 = -8$ and $\beta_1 = 660$.
   (b) $\beta_0 = 652$ and $\beta_1 = 8$.
   (c) $\beta_0 = 660$ and $\beta_1 = 652$.
   (d) $\beta_0 = 660$ and $\beta_1 = -8$.

10. In a wage–education regression, $E[u_i \mid \text{Education}_i] = 0$, but $\text{Var}(u_i \mid \text{Education}_i = x)$ is larger at higher education levels. This describes

    (a) homoskedasticity, because the conditional error mean is zero.
    (b) heteroskedasticity, which can coexist with A1.
    (c) a requirement that every worker have the same wage.
    (d) a necessary violation of A1.

**2. True or False Questions (20 points total; 4 questions; 5 points each)**

**Instructions:** On your separate answer sheets, label each part and write **T** or **F**. You must justify each answer. Full credit requires a correct answer and justification.

(a) Given A1–A3, variation in $X$, and a sufficiently large sample, heteroskedasticity-robust standard errors are valid under both homoskedasticity and heteroskedasticity. Requesting robust standard errors leaves the OLS coefficient estimates unchanged.

(b) In a large-sample two-sided test of a zero test-score–income slope, rejecting at the 5% significance level implies that the same result must also reject at the 1% significance level.

(c) A researcher finds that OLS residuals average to zero and are uncorrelated with the regressor in the sample. These facts prove A1, $E[u_i \mid X_i] = 0$, for the population model.

(d) The standard error of the estimated wage–education slope estimates the standard deviation of slope estimates across repeated random samples of the same size. The reported standard error itself generally changes from sample to sample.

### 3. Analytical Question (50 points total)

**Instructions:** Write your answers and show all work on your separate answer sheets. Label each part (a)–(g). Interpret coefficients using words and units. Use the displayed two-decimal coefficients and standard errors for all calculations; report every calculated answer to two decimal places.

A researcher uses data on 420 California school districts. The outcome `testscr` is the district's average test score (in points), and `avginc` is average district income (in **thousands of dollars**). Observed income ranges from approximately 5.34 to 55.33 thousand dollars. The researcher estimates

$$\text{TestScore}_i = \beta_0 + \beta_1 \text{Income}_i + u_i$$

using `regress testscr avginc, robust`. The Stata coefficient table, rounded to two decimal places, is:

```
Linear regression                               Number of obs = 420

------------------------------------------------------------------------------
             |               Robust
     testscr | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      avginc |       1.88       0.11    16.53    0.00         1.66        2.10
       _cons |     625.38       1.87   334.81    0.00       621.71      629.06
------------------------------------------------------------------------------
```

(a) (6 points) Write the sample regression equation, place each coefficient's robust standard error in parentheses below it, and report the sample size.

(b) (6 points) Interpret the income slope. Compute the predicted score for income of \$20,000 and the predicted score change associated with an income increase of \$5,000.

(c) (4 points) Interpret the intercept. Explain why its practical interpretation is limited in this dataset.

(d) (6 points) Explain what the two reported coefficient standard errors estimate. Calculate the income slope's $t$-statistic for testing a zero slope, and explain what its magnitude means.

**Analytical Question (continued)**

Continue using the same regression output on the previous page. For calculations here, use the handout's large-sample normal critical values: 1.65 (90%), 1.96 (95%), and 2.58 (99%). Stata computes its reported statistics before display rounding. Recomputed statistics and intervals using the displayed coefficients and SEs can therefore differ slightly; use the rounded values for your calculations.

(e) (8 points) State the null and two-sided alternative for testing a zero income slope. At the 5% significance level, use each of the three rejection rules separately: the reported 95% CI, the $t$-statistic, and the $p$-value. Read and interpret the reported income CI. Does the displayed $p$-value of `0.00` in Stata output mean that the probability is exactly zero?

(f) (8 points) Compute the 90% and 99% normal-approximation confidence intervals for the income slope. Compare their centers and widths. Is the slope statistically different from zero at the 10%, 5%, and 1% significance levels? Explain how changing confidence or significance levels changes the decision rule, even if the decision for this empirical example stays the same.

(g) (12 points total) Evaluate the assumptions for this observational regression. Label your answers (g)(i)–(g)(iv).

(i) (4 points) State A1. Give one omitted determinant of test scores that might threaten A1 for a causal income effect, and explain why. Can the sample residuals establish A1?

(ii) (3 points) State A2. Discuss what would make it plausible or doubtful for these districts. Can the regression table establish A2?

(iii) (2 points) State A3. What would you examine to assess its plausibility? Can the regression table establish A3?

(iv) (3 points) Explain whether the `robust` option establishes homoskedasticity, repairs a failure of A1, or resolves dependence between districts.

**4. Bonus Question (5 points)**

**Instructions:** Answer on your separate answer sheets. Label parts (a) and (b), and explain your reasoning.

Compare two hypothetical wage–education studies. Assume A1–A3, homoskedasticity, positive education variance, the same measurement units, and sufficiently large samples in both studies. Relative to Study A, Study B has twice the sample size, twice the population regression-error variance, and four times the population variance of education.

(a) (3 points) Using the homoskedastic large-sample slope-variance formula, calculate the ratio of Study B's slope variance to Study A's. Also calculate the ratio of their slope standard deviations (and thus the approximate ratio of typical reported standard errors).

(b) (2 points) Now suppose A1 fails in Study B because ability affects wages and is systematically related to education. Would the smaller sampling uncertainty calculated in part (a) establish that Study B's slope estimates the causal effect of education? Explain.